

THEORY AND METHODS

A MAXIMUM LIKELIHOOD ESTIMATION PROCEDURE FOR THE GENERALIZED LINEAR MODEL WITH RESTRICTIONS

G.B. Matthews

University of Natal, Durban, 4001, South Africa.

and

N.A.S. Crowther

Department of Statistics, University of Pretoria, Pretoria, 0002, South Africa.

Key words: Constraints; generalized linear models; logistic regression; maximum likelihood estimation; Poisson regression; splines.

Summary: A maximum likelihood (ML) estimation procedure is presented for estimating the parameter vector in the generalized linear model in the presence of linear constraints on the elements of the vector of parameters. Matthews and Crowther (1995) propose a ML estimation procedure which gives the ML estimate of the mean vector of the multivariate exponential distribution subject to constraints on the mean vector. An application of the estimation procedure provides a simple approach to the problem of constrained ML estimation in the generalized linear model. The strength of the procedure is emphasized in a logistic regression example, where certain constraints are placed on the regression parameters. A further application to cubic splines is discussed in the context of the constrained estimation framework.

1. Introduction

Generalized linear models were first introduced by Nelder and Wedderburn (1972), and this class of models has subsequently received a great deal of

attention. Using the notation of McCullagh and Nelder (1989), independent observations $y_1, \dots, y_n$ are made on a response variable Y , whose probability distribution belongs to the exponential family. A monotonic, differentiable function, $h(\cdot)$, of the mean is then expressed as a linear model, i.e. $h(\mu) = \mathbf{X}\beta$, where $E(y) = \mu$, $\mathbf{X}$ is an appropriate matrix of observations on the covariates and β is a vector of unknown parameters. The ML estimates of β are usually found by means of the iteratively reweighted least squares (IRLS) procedure or the Newton-Raphson procedure.

The purpose of this paper is to present a ML estimation procedure which can be used to find the ML estimate of β in the model $h(\mu) = \mathbf{X}\beta$ under a set of linear restrictions $\mathbf{L}\beta = \mathbf{d}$, where $\mathbf{d}$ is a vector of constants. We thus require the ML estimate of β subject to the constraints $\mathbf{L}\beta = \mathbf{d}$. Denote this constrained ML estimate by $\hat{\beta}_c$.

The problem of constrained ML estimation is not new and one of the earliest references to this problem in a more general setting is that of Aitchison and Silvey (1958). Restricted estimation in the generalized linear model, which is the topic of this paper, has been discussed by Nyquist (1991), where the quadratic penalty function approach is used to find the restricted ML estimate, $\hat{\beta}_c$.

In this paper we apply a ML estimation procedure proposed by Matthews and Crowther (1995), to find the ML estimate, $\hat{\beta}_c$. The procedure in the latter reference provides the ML estimate of the mean vector of the multivariate exponential family, when modelling in terms of certain constraints on the elements of the mean vector. An application of this procedure provides a simple solution to the problem of restricted estimation in the generalized linear model. A further advantage is that the procedure incorporates all distributions that

satisfy the definition of the probability function of the multivariate exponential family. This will include the likelihood function of a random sample drawn from the univariate exponential family.

In section 2 we introduce some notation, define the probability function of the multivariate exponential family and present the estimation procedure of Matthews and Crowther (1995). The work in section 3 provides a general expression for the constrained ML estimate, $\hat{\beta}_c$ in the generalized linear model. In section 4 we consider the logistic regression model with constraints and present an example to illustrate the procedure. The fitting of a cubic spline to normal data is presented in section 5, as a further example of the constrained normal procedure. In section 6, we show how the estimation procedure can be applied to a regression problem that requires a perfect fit at one or more of the points, or in other words, the regression model must pass through one or more pre-specified points.

2. A maximum likelihood estimation procedure

First consider the definition of the probability function of the multivariate exponential family. Let y be a $p \times 1$ random vector and θ be a $p \times 1$ vector of parameters. The probability function for the exponential family is given by

$$\begin{aligned} p(y, \theta) &= a(\theta)b(y) \exp(y' \theta), \quad y \in \mathbb{R}^p, \quad \theta \in \mathcal{N} \\ &= b(y) \exp \{y' \theta - \kappa(\theta)\}, \end{aligned} \tag{1}$$

where $\kappa(\theta) = -\ln a(\theta)$ and $\mathcal{N}$ is the natural parameter space for the canonical parameter θ . The mean vector and covariance matrix of y are

given by

$$E(y) = \mu = \frac{\partial}{\partial \theta} \kappa(\theta) \text{ and } Cov(y) = \frac{\partial^2 \kappa(\theta)}{\partial \theta \partial \theta'} = \left\{ \frac{\partial^2 \kappa(\theta)}{\partial \theta_i \partial \theta_j} \right\} = V_\mu$$

We shall require the following proposition which is proved in the reference Matthews and Crowther (1995). The proposition provides the ML estimation procedure for the mean vector of the multivariate exponential distribution, subject to certain constraints on the mean vector and an application of the proposition provides the constrained ML estimate $\hat{\beta}_c$ for the generalized linear model.

PROPOSITION 1

Consider a random vector y , with probability function belonging to the exponential family. Let $g(\mu)$ be a continuous vector valued function of μ , for which the first order partial derivatives exist. Let $G_\mu = \frac{\partial g(\mu)}{\partial \mu}$ be the derivative of $g(\mu)$ with respect to μ and $G_y = \frac{\partial g(\mu)}{\partial \mu} \Big|_{\mu=y}$.

The ML estimate of μ subject to the constraints $g(\mu) = 0$, is given by

$$\hat{\mu}_c = y - (G_\mu V_\mu)' (G_y V_\mu G_\mu')^{-1} g(y) + o(\|y - \mu\|). \quad (2)$$

This result implies that the ML estimate must in general, be obtained iteratively.

In the proof the ML estimate is found by maximizing the probability function in (1) subject to the given constraints, $g(\mu) = 0$, by using a vector of Lagrange multipliers and a Taylor series expansion.

It was previously implicitly assumed that the restrictions $g(\mu)$ are linearly independent. If this is not the case, assume that

$$g^0(\mu) = \begin{pmatrix} B_1 \\ B_2 \end{pmatrix} g(\mu) = Bg(\mu), \text{ where } B = \begin{pmatrix} B_1 \\ B_2 \end{pmatrix}$$

for constant matrices B_1 and B_2 . The matrix B_1 is non-singular and the elements of $g(\mu)$ are linearly independent. then

$$G_\mu^0 = \frac{\partial g^0(\mu)}{\partial \mu} = \begin{pmatrix} B_1 \\ B_2 \end{pmatrix} G_\mu \text{ and } G_y^0 = \frac{\partial g^0(\mu)}{\partial \mu} \Big|_{\mu=y} = \begin{pmatrix} B_1 \\ B_2 \end{pmatrix} G_y.$$

Since the matrix $B'(BDB')^*B$ is independent of the choice of the generalized inverse $(BDB')^*$ of (BDB') for any non-singular D , the expression

$$\begin{aligned} \hat{\mu}_c &= y - (G_\mu^0 V_\mu)' (G_y^0 V_\mu G_\mu^{0'})^* g^0(y) + o(\|y - \mu\|) \\ &= y - (BG_\mu V_\mu)' (BG_y V_\mu G_\mu' B')^* Bg(y) + o(\|y - \mu\|) \end{aligned}$$

is independent of the choice of the generalized inverse. Since

$$\begin{pmatrix} B_1'^{-1} \\ 0 \end{pmatrix} (G_y V_\mu G_\mu')^{-1} (B_1^{-1}, 0)$$

is also a generalized inverse of

$$\begin{pmatrix} B_1 \\ B_2 \end{pmatrix} G_y V_\mu G_\mu' (B_1', B_2')$$

it follows that

$$(G_\mu^0 V_\mu)' (G_y^0 V_\mu G_\mu^{0'})^* g^0(y) = (G_\mu V_\mu)' (G_y V_\mu G_\mu')^{-1} g(y).$$

This proves the fact that if the restrictions $g(\mu)$ are linearly dependent, Proposition 1 still holds except for the fact that the inverse is to be replaced by any generalized inverse.

It is also useful to have the covariance matrix of the ML estimate, $\hat{\mu}_c$ and this is given in the next proposition.

PROPOSITION 2

The asymptotic covariance matrix of $\hat{\mu}_c$ is given by

$$\Sigma_c = V_\mu - (G_\mu V_\mu)' (G_\mu V_\mu G_\mu')^{-1} G_\mu V_\mu.$$

The proof of this result also appears in Matthews and Crowther (1995). If the restrictions $g(\mu)$ are linearly dependent, it follows as in the case of Proposition 1 that the inverse should be replaced by any generalized inverse.

We now show how Proposition 1 may be implemented to find the ML estimate in the generalized linear model subject to restrictions placed on the parameter vector β .

3. Restricted estimation in the generalized linear model

Consider the vector random variable y with pdf belonging to the exponential family, with mean $E(y) = \mu$ and variance covariance matrix $Cov(y) = V_\mu$.

The objective is to find the ML estimate $\hat{\beta}_c$ in the generalized linear model $h(\mu) = X\beta$, with a set of linear restrictions $L\beta = d$ imposed on β . The restrictions should be well defined (estimable) in the sense that $L = L_1X$ for some L_1 .

We will now write the model $h(\mu) = X\beta$ along with the restrictions $L\beta = d$ in the form $g(\mu) = 0$ in order to implement Proposition 1. To do this let $P = I - X(X'X)^*X'$ and $A = (X'X)^*X'$. Then $Ph(\mu) = PX\beta = 0$ and $LAh(\mu) = L_1X(X'X)^*X'X\beta = L\beta$. Hence $LAh(\mu) - d = 0$, since $L\beta = d$. We now have a function

$$g(\mu) = \begin{pmatrix} Ph(\mu) \\ LAh(\mu) - d \end{pmatrix} = \begin{pmatrix} Ph(\mu) \\ Ch(\mu) - d \end{pmatrix} = 0, \quad (3)$$

where $C = LA$.

It follows that

$$G_\mu = \frac{\partial g(\mu)}{\partial \mu} = \begin{pmatrix} P \\ C \end{pmatrix} \frac{\partial h(\mu)}{\partial \mu} = M \frac{\partial h(\mu)}{\partial \mu}, \quad (4)$$

where $M = \begin{pmatrix} P \\ C \end{pmatrix}$. From Proposition 1, the ML estimate of the mean vector subject to the constraints, is given by

$$\hat{\mu}_c = y - (G_{\mu} V_{\mu})' (G_y V_{\mu} G_{\mu}')^* g(y) + o(\|y - \mu\|), \quad (5)$$

which in general implies an iterative procedure.

If $X'X$ is non-singular, then $(X'X)^{-1}$ exists and $A = (X'X)^{-1}X'$. Furthermore, since $h(\hat{\mu}_c) = X\hat{\beta}_c$, it follows that $\hat{\beta}_c = A h(\hat{\mu}_c)$ and is unique. This expression gives the constrained ML estimate $\hat{\beta}_c$, which is found by using the ML estimate $\hat{\mu}_c$, found in (5), on convergence of the iterative procedure.

We are also able to find the asymptotic covariance matrix of $\hat{\beta}_c$.

Let

$$H = \frac{\partial h(\mu)}{\partial \mu} \text{ and } H_{\hat{\mu}_c} = \frac{\partial h(\mu)}{\partial \mu} \Big|_{\mu=\hat{\mu}_c}.$$

Then the asymptotic covariance matrix of $h(\hat{\mu}_c)$ is $H\Sigma_c H'$ and it follows that the asymptotic covariance matrix of $\hat{\beta}_c = A h(\hat{\mu}_c)$ is $AH\Sigma_c H' A'$. The estimated asymptotic covariance matrix of $\hat{\beta}_c$ is $AH_{\hat{\mu}_c} \hat{\Sigma}_c H'_{\hat{\mu}_c} A'$, where

$$\hat{\Sigma}_c = V_{\hat{\mu}_c} - (G_{\hat{\mu}_c} V_{\hat{\mu}_c})' (G_{\hat{\mu}_c} V_{\hat{\mu}_c} G'_{\hat{\mu}_c})^* G_{\hat{\mu}_c} V_{\hat{\mu}_c}.$$

The estimated covariance matrix thus uses the ML estimate $\hat{\mu}_c$, which is obtained on convergence of the iterative procedure in (5).

These results can now be implemented for distributions satisfying the definition of the multivariate exponential family. It can be shown that if $y = (y_1, \dots, y_k)'$, where for example, the y_i are k independent observations from binomial distributions with parameters n_i and π_i , then the pdf of y , which is the joint pdf of the k independent binomial random variables, can be written in the form of equation (1). Similarly k independent random

variables from a normal distribution, or a Poisson distribution, will also have a pdf satisfying the definition in equation (1).

In the next section we show how the procedure may be implemented for a logistic regression model with constraints on the vector of parameters and this is illustrated by an example.

4. Logistic regression with restrictions

Let Y_i , $i = 1, \dots, k$ be independent random variables, where Y_i has a binomial distribution with parameters n_i and π_i . Suppose that the covariates $X_1, \dots, X_m$ are observed and that for occasion i , $x_i = (x_{1i}, x_{2i}, \dots, x_{mi})'$ and y_i is the number of successes in the n_i trials, $i = 1, \dots, k$. The logistic regression model is

$$\eta_i = \log \left(\frac{\pi_i}{1 - \pi_i} \right) = \beta_0 + \beta_1 x_{1i} + \beta_2 x_{2i} + \dots + \beta_m x_{mi}, \quad i = 1, \dots, k.$$

This model may be written as

$$\eta = h(\mu) = \mathbf{X}\beta,$$

where $\mu = (\mu_1, \dots, \mu_k)'$, $\mu_i = E(Y_i) = n_i \pi_i$ and $\beta = (\beta_0, \dots, \beta_m)'$.

We may write

$$\eta_i = \log \left(\frac{\mu_i}{n_i - \mu_i} \right)$$

so that

$$\frac{\partial \eta_i}{\partial \mu_i} = \frac{1}{n_i \pi_i (1 - \pi_i)}, \quad i = 1, \dots, k.$$

Let $\mathbf{V}_\mu = \text{diag} [n_i \pi_i (1 - \pi_i)]$. Then

$$\frac{\partial \eta}{\partial \mu} = \frac{\partial h(\mu)}{\partial \mu} = \mathbf{V}_\mu^{-1}.$$

Following the notation of section 3, we now have $\mathbf{P}\eta = \mathbf{P}h(\mu) = \mathbf{P}\mathbf{X}\beta = 0$ and the imposed restrictions $\mathbf{L}\beta = \mathbf{d}$. Thus the model with the restrictions may be written in the form

$$\mathbf{g}(\mu) = \begin{pmatrix} \mathbf{P}h(\mu) \\ \mathbf{C}h(\mu) - \mathbf{d} \end{pmatrix} = 0.$$

Applying equation (4) of section 3, we have

$$\mathbf{G}_\mu = \frac{\partial \mathbf{g}(\mu)}{\partial \mu} = \mathbf{M} \frac{\partial h(\mu)}{\partial \mu} = \mathbf{M}\mathbf{V}_\mu^{-1}.$$

The ML estimate of the mean vector subject to the constraints is given by

$$\begin{aligned} \hat{\mu}_c &= \mathbf{y} - (\mathbf{G}_\mu \mathbf{V}_\mu)' (\mathbf{G}_y \mathbf{V}_\mu \mathbf{G}_\mu')^* \mathbf{g}(\mathbf{y}) + o(\|\mathbf{y} - \mu\|) \\ &= \mathbf{y} - \mathbf{M}' (\mathbf{M}\mathbf{V}_\mu^{-1} \mathbf{M}')^* \mathbf{g}(\mathbf{y}) + o(\|\mathbf{y} - \mu\|) \end{aligned} \quad (6)$$

which is obtained by iteration. In this case the iterative procedure will run as follows. Let $\mu^{(r)}$ denote the r th approximation for the ML estimate $\hat{\mu}_c$. At step r in the process ($r = 0, 1, 2, \dots$),

$$\mu^{(r+1)} = \mu^{(r)} - \mathbf{M}' (\mathbf{M}\mathbf{V}_{\mu^{(r)}}^{-1} \mathbf{M}')^* \mathbf{g}(\mu^{(r)}),$$

where for $r = 0$, $\mu^{(0)} = \mathbf{y}$, the original data vector. This process converges to the ML estimate $\hat{\mu}_c$. We then calculate $\hat{\beta}_c = \mathbf{A}h(\hat{\mu}_c)$, where from section 3, $\mathbf{A} = (\mathbf{X}'\mathbf{X})^{-1}\mathbf{X}'$.

The next example illustrates a logistic regression model with constraints on β .

Example 1: Toxicity of insecticides to flour beetles

In an insecticidal trial reported by Hewlett and Plackett (1950), flour beetles, *Tribolium castaneum*, were sprayed with one of three different insecticides in solution in Shell oil P31. The three insecticides used were dichlorodiphenyltrichloroethane (DDT) at 2.0% w/v, γ -benzene hexachloride (γ -BHC) used at 1.5% w/v and a mixture of the two. In the experiment, batches

of about fifty insects were exposed to varying deposits of spray, measured in units of $\text{mg}/10 \text{ cm}^2$. The resulting data on the proportion of insects killed after a period of six days are given in the table below.

Table 1
Toxicity to the flour beetle of sprays of DDT, γ -BHC and the two together

Insecticide	Deposit of insecticide					
	2.00	2.64	3.48	4.59	6.06	8.00
DDT	3/50	5/49	19/47	19/38	24/49	35/50
γ -BHC	2/50	14/49	20/50	27/50	41/50	40/50
DDT + γ -BHC	28/50	37/50	46/50	48/50	48/50	50/50

In modelling these data, the logarithm of the amount of deposit of insecticide will be used as the explanatory variable in a linear logistic model. The first step is to look at the extent to which there are differences between the insecticides in terms of the relationship between the probability of death and the amount of deposit.

For convenience of further discussions, refer to insecticides DDT, γ -BHC and DDT + γ -BHC as treatments 1, 2 and 3 respectively. Consider the logistic regression model

$$\text{MODEL 1} : \eta_{ij} = \ln \left(\frac{p_{ij}}{1 - p_{ij}} \right) = \beta_{0i} + \beta_{1i} \ln(x_j);$$

$$i = 1, 2, 3; \quad j = 1, \dots, 6;$$

where p_{ij} is the proportion of insects killed, applying treatment i at dosage level x_j of the insecticide. This model may be written as $\eta = h(\mu) = X\beta$,

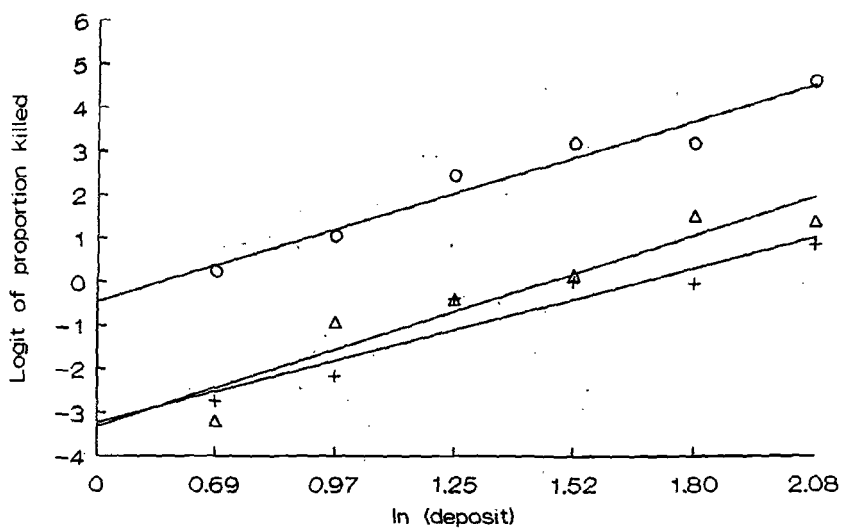


Figure 1. Plot of the empirical logit of the proportion of insects killed against $\ln(\text{deposit})$ for DDT (+), γ -BHC (Δ), and DDT + γ -BHC (o)

where $\mu = (\mu_{11}, \mu_{12}, \dots, \mu_{36})'$, $\beta = (\beta_{01}, \beta_{02}, \beta_{03}, \beta_{11}, \beta_{12}, \beta_{13})'$
and

$$X = \begin{pmatrix} 1 & 0 & 0 & x_1 & 0 & 0 \\ 1 & 0 & 0 & x_2 & 0 & 0 \\ \vdots & \vdots & & & & \vdots \\ 1 & 0 & 0 & x_6 & 0 & 0 \\ 0 & 1 & 0 & 0 & x_1 & 0 \\ 0 & 1 & 0 & 0 & x_2 & 0 \\ \vdots & \vdots & & & & \vdots \\ 0 & 1 & 0 & 0 & x_6 & 0 \\ 0 & 0 & 1 & 0 & 0 & x_1 \\ 0 & 0 & 1 & 0 & 0 & x_2 \\ \vdots & \vdots & & & & \vdots \\ 0 & 0 & 1 & 0 & 0 & x_6 \end{pmatrix}$$

This model implies 3 regression lines, each with a different slope and intercept. The parameter estimates for this model are:

$$\begin{aligned} \hat{\beta}_{01} &= -3.803 \text{ (s.e.} = 0.5002), & \hat{\beta}_{02} &= -4.0428 \text{ (s.e.} = 0.4972), \\ \hat{\beta}_{03} &= -2.1207 \text{ (s.e.} = 0.5908), & \hat{\beta}_{11} &= 2.2824 \text{ (s.e.} = 0.3190), \\ \hat{\beta}_{12} &= 2.8381 \text{ (s.e.} = 0.3392), & \hat{\beta}_{13} &= 3.3884 \text{ (s.e.} = 0.5733), \end{aligned}$$

The measure of goodness of fit used for generalized linear models is the deviance. The deviance for k independent binomial populations is given by

$$D = 2 \sum_{i=1}^k [y_i \ln(y_i/\hat{\mu}_i) + (n_i - y_i) \ln((n_i - y_i)/(n_i - \hat{\mu}_i))],$$

and has an asymptotic χ^2 distribution, (McCullagh and Nelder (1989)).

The deviance for Model 1, $D = 19.29$ with $df = 12$.

Let us now consider Model 2, having 3 lines parallel but with different intercepts, i.e. $\beta_{11} = \beta_{12} = \beta_{13}$. These constraints can be written in the

form $L\beta = 0$, where

$$L = \begin{pmatrix} 0 & 0 & 0 & 1 & -1 & 0 \\ 0 & 0 & 0 & 0 & 1 & -1 \end{pmatrix}$$

Applying the estimation procedure discussed earlier, gives

$$\begin{aligned} \hat{\beta}_{01} &= -4.4541 \text{ (s.e.} = 0.3575), & \hat{\beta}_{02} &= -3.8397 \text{ (s.e.} = 0.3313), \\ \hat{\beta}_{03} &= -1.4227 \text{ (s.e.} = 0.2842), & \hat{\beta}_{11} = \hat{\beta}_{12} = \hat{\beta}_{13} &= 2.6938 \\ & & & \text{(s.e.} = 0.2146). \end{aligned}$$

The deviance for the parallel lines model is $D = 22.69$ with $df = 14$.

A closer look at figure 1 shows that insecticides γ -BHC(Δ) and DDT + γ -BHC(o) seem to have parallel slopes, while DDT(+) and γ -BHC(Δ) show similar intercepts. We thus consider Model 3, with slopes, $\beta_{12} = \beta_{13}$ and intercepts, $\beta_{01} = \beta_{02}$. These constraints are also of the form $L\beta = 0$, where

$$L = \begin{pmatrix} 0 & 0 & 0 & 0 & 1 & -1 \\ 1 & -1 & 0 & 0 & 0 & 0 \end{pmatrix}.$$

Once again applying the estimation procedure with L above, we obtain the ML estimates

$$\begin{aligned} \hat{\beta}_{01} = \hat{\beta}_{02} &= -4.0830 \text{ (s.e.} = 0.3278), & \hat{\beta}_{03} &= -1.6134 \text{ (s.e.} = 0.2934), \\ \hat{\beta}_{11} &= -2.4367 \text{ (s.e.} = 0.2192), & \hat{\beta}_{12} = \hat{\beta}_{13} &= -2.8798 \\ & & & \text{(s.e.} = 0.2282). \end{aligned}$$

The deviance for this model, $D = 20.42$ with $df = 14$. We note that this model has a smaller deviance than the parallel lines model and both models have $df = 14$. The deviance $D = 20.42$ is also not much greater than the deviance for Model 1, having different slopes and intercepts ($D = 19.29$, $df = 12$). For the gain in degrees of freedom, it is thus not unreasonable to fit Model 3 having the constraints $\beta_{12} = \beta_{13}$ and $\beta_{01} = \beta_{02}$. This means that insecticides γ -BHC (treatment 2) and DDT + γ -BHC (treatment 3) have the same slopes. The bioassay interpretation for these two insecticides is as follows: The relative potency of two compounds is defined to be the ratio of equally effective doses. Suppose that dose d_s of the standard procedure

and dose d_n of the new drug produce the same result, say λ on the logit scale, so that these two drugs are equally potent; then the potency of the new drug relative to the standard is $r_{ns} = d_s/d_n$.

In our example, equal logits for doses d_2 and d_3 of treatments 2 and 3, respectively, implies that

$$\hat{\beta}_{02} + \hat{\beta}_{12} \ln(d_2) = \hat{\beta}_{03} + \hat{\beta}_{13} \ln(d_3)$$

This gives

$$r_{32} = d_2/d_3 = \exp \left[(\hat{\beta}_{03} - \hat{\beta}_{02}) / \hat{\beta}_* \right]$$

where $\hat{\beta}_* = \hat{\beta}_{12} = \hat{\beta}_{13}$. Substituting the estimates into this expression gives $r_{32} = \exp(0.8576) = 2.36$. This means that DDT + γ -BHC is approximately 2.4 times more effective than γ -BHC alone.

The estimated parameters for these constrained models are easily computed using a matrix algebra package, for example the SAS/IML procedure of the SAS System. The SAS/IML program for this example appears in the appendix, where the matrix L corresponds to the matrix appropriate for the constrained model being fitted. It should also be noted that the estimates for Model 1, with no constraints on the parameters can be found by taking $L = 0_{6 \times 6}$.

5. Restricted regression for normal data

Another useful application is the fitting of a cubic spline to data coming from a normal distribution. Consider the general linear model, for which the so-called link function is the identity, namely $E(y) = \mu = X\beta$ and $Cov(y) = \sigma^2 I = V$, with further constraints $L\beta = d$. In this case, in terms of the notation used earlier, $h(\mu) = \mu$ and $g(\mu) = \begin{pmatrix} P\mu \\ C\mu - d \end{pmatrix} = 0$. It

also then follows that

$$\mathbf{G}_\mu = \mathbf{G} = \begin{pmatrix} \mathbf{P} \\ \mathbf{C} \end{pmatrix}.$$

The ML estimate of the mean vector subject to the restrictions, is

$$\hat{\mu}_c = \mathbf{y} - (\mathbf{G}\mathbf{V})'(\mathbf{G}\mathbf{V}\mathbf{G}')^* \mathbf{G}\mathbf{y}. \quad (7)$$

For $\mathbf{V} = \sigma^2 \mathbf{I}$, the ML estimate is $\hat{\mu}_c = \mathbf{y} - \mathbf{G}'(\mathbf{G}'\mathbf{G})^* \mathbf{G}\mathbf{y}$, which is found without iteration.

The following example in which 3 cubic spline functions are fitted over a certain region, illustrates the procedure.

Consider a regression problem where the domain of the predictor variable x , can be sub-divided into 3 regions and over each region a cubic function will be fitted. The regression equation is of the form

$$\begin{aligned} E(y_i) = & \beta_{01} + \beta_{11}x_{1i} + \beta_{21}x_{1i}^2 + \beta_{31}x_{1i}^3 \\ & + \beta_{02} + \beta_{12}x_{2i} + \beta_{22}x_{2i}^2 + \beta_{32}x_{2i}^3 \\ & + \beta_{03} + \beta_{13}x_{3i} + \beta_{23}x_{3i}^2 + \beta_{33}x_{3i}^3, \quad i = 1, \dots, n. \end{aligned} \quad (8)$$

If an x -value falls in region 1, then the variable x_1 will take on this value and the variables x_2 and x_3 will be zero. Similarly for x -values in regions 2 and 3. Suppose further that there are 2 change points or knots, namely at $x = c_1$ and $x = c_2$, which are points that are part of the data set. The model in (8) can be written as

$$\mu = E(y) = \mathbf{X}\boldsymbol{\beta},$$

where $\boldsymbol{\beta}' = (\beta_{01}, \beta_{11}, \dots, \beta_{31}, \dots, \beta_{03}, \dots, \beta_{33})$, and $\mathbf{X} : n \times 12$ is the appropriate matrix of observations on the covariates.

At the first knot $x = c_1$, the cubic functions for regions 1 and 2 must be equal, i.e.

$$\beta_{01} + \beta_{11}c_1 + \beta_{21}c_1^2 + \beta_{31}c_1^3 = \beta_{02} + \beta_{12}c_1 + \beta_{22}c_1^2 + \beta_{32}c_1^3$$

or

$(1, c_1, c_1^2, c_1^3, 0, \dots, 0, \dots, 0)\beta = (0, \dots, 0, 1, c_1, c_1^2, c_1^3)\beta$,
which can be written as $l'_1\beta = 0$, where $l'_1 = (1, c_1, c_1^2, c_1^3, -1, -c_1, -c_1^2, -c_1^3, 0, 0, 0, 0)$. Similarly at $x = c_2$ the second two cubic functions must be equal, i.e.

$$\beta_{02} + \beta_{12}c_2 + \beta_{22}c_2^2 + \beta_{32}c_2^3 = \beta_{03} + \beta_{13}c_2 + \beta_{23}c_2^2 + \beta_{33}c_2^3,$$

which may be written in the form $l'_2\beta = 0$.

Furthermore at the first knot, $x = c_1$, the derivatives of the two adjacent functions must be equal. Firstly

$$\frac{\partial}{\partial x_{1i}} E(y_i) = \beta_{11} + 2\beta_{21}x_{1i} + 3\beta_{31}x_{1i}^2 \quad (9)$$

and

$$\frac{\partial}{\partial x_{2i}} E(y_i) = \beta_{12} + 2\beta_{22}x_{2i} + 3\beta_{32}x_{2i}^2. \quad (10)$$

Equations (9) and (10) must be equal for $x_{1i} = x_{2i} = c_1$ therefore

$$\beta_{11} + 2\beta_{21}c_1 + 3\beta_{31}c_1^2 = \beta_{12} + 2\beta_{22}c_1 + 3\beta_{32}c_1^2,$$

or

$$(0, 1, 2c_1, 3c_1^2, 0, -1, -2c_1, -3c_1^2, 0, 0, 0, 0)\beta = 0,$$

which may be written as $l'_3\beta = 0$.

Similarly at knot $x = c_2$, the derivatives of the next two adjacent functions must be equal, which will give

$$(0, 0, 0, 0, 0, 1, 2c_2, 3c_2^2, 0, -1, -2c_2, -3c_2^2)\beta = 0, \text{ or } l'_4\beta = 0.$$

The four constraints $l'_i\beta = 0$, $i = 1, \dots, 4$, may now be written as

$$L\beta = 0, \text{ where } L = (l'_1, l'_2, l'_3, l'_4)'.$$

This problem is a general linear model, $\mu = E(y) = X\beta$ subject to the constraints $L\beta = 0$. Equation (7) will give the ML estimate $\hat{\mu}_c$.

Example 2

The data in Table 2, taken from Montgomery and Peck (1992), are the voltage drop in a guided missile motor y , observed over the time of the missile flight x (in seconds). A scatter plot suggests that voltage drop behaves differently in different segments of time. The data is thus modelled by a cubic spline using two knots at $c_1 = 6.5$ and $c_2 = 13$ seconds after launching.

Table 2
Battery voltage drop (y) over time (x) for guided missile motor

x	y	μ_c
0.0	8.33	8.34
0.5	8.23	7.85
1.0	7.17	7.54
1.5	7.14	7.38
2.0	7.31	7.38
2.5	7.60	7.51
3.0	7.94	7.76
3.5	8.30	8.11
4.0	8.76	8.55
4.5	8.71	9.06
5.0	9.71	9.64
5.5	10.26	10.26
6.0	10.91	10.92
6.5	11.67	11.59
7.0	11.76	12.23
7.5	12.81	12.79
8.0	13.30	13.29
8.5	13.88	13.71
9.0	14.59	14.06
9.5	14.05	14.35

10.0	14.48	14.56
10.5	14.92	14.70
11.0	14.37	14.77
11.5	14.63	14.77
12.0	15.18	14.71
12.5	14.52	14.57
13.0	14.34	14.36
13.5	13.81	14.08
14.0	13.79	13.72
14.5	13.05	13.30
15.0	13.04	12.84
15.5	12.60	12.35
16.0	12.05	11.86
16.5	11.14	11.37
17.0	11.15	10.92
17.5	10.14	10.50
18.0	10.08	10.15
18.5	9.78	9.88
19.0	9.80	9.70
19.5	9.95	9.63
20.0	9.51	9.70

Applying the estimation procedure in (7), we find the vector of fitted values, $\hat{\mu}_c$, which is given in Table 2.

The error sum of squares, $SSE = (\hat{\mu}_c - y)'(\hat{\mu}_c - y) = 2.3552$, the regression sum of squares, $SSR = 260.3333$ and the coefficient of multiple determination for the model, $R^2 = 0.99$, indicating that the fit is very good.

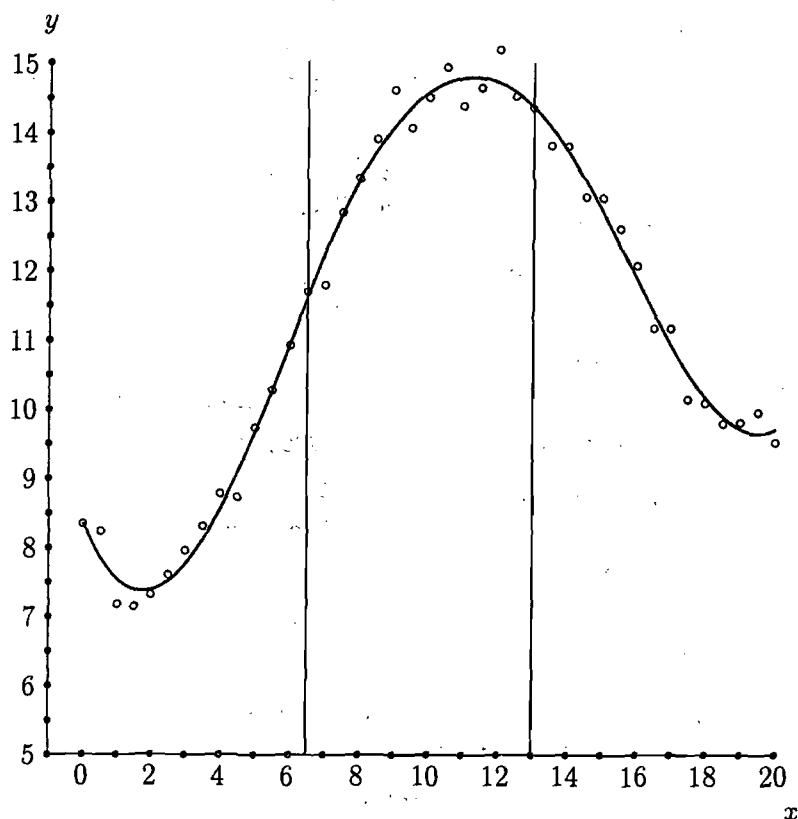


Figure 2. Battery voltage drop over time (Montgomery & Peck, 1992).

6. Regression through a fixed point

A generalized regression model with the constraint that there must be a perfect fit at a certain fixed point, also fits into the estimation framework of section

3. Suppose that in fitting the generalized linear model $h(\mu) = X\beta$, there is a further constraint that the model must pass through the point $(h(y_0), x'_0)$, i.e. $h(y_0) = x'_0\beta$, where $x'_0 = (1, x_{10}, \dots, x_{k0})$. This can be written as the constraint $x'_0\beta - h(y_0) = 0$. Using the notation of section 3, the function $g(\mu)$ in equation (3) becomes

$$g(\mu) = \begin{pmatrix} Ph(\mu) \\ x'_0Ah(\mu) - h(y_0) \end{pmatrix} = 0.$$

It then follows that

$$G_\mu = \frac{\partial g(\mu)}{\partial \mu} = \begin{pmatrix} P \\ x'_0A \end{pmatrix} \frac{\partial h(\mu)}{\partial \mu}.$$

The iterative procedure given in equation (2) may now be applied, with $g(\mu)$ and G_μ above, to find the ML estimate $\hat{\mu}_c$. This iterative procedure can be applied to any set of observations y coming from a multivariate exponential distribution and for any link function $h(\cdot)$. It is also obvious that the above-mentioned technique can be extended to cater for a model with a perfect fit in more than one point.

7. Discussion

Two other applications warranting discussion are the restricted Poisson regression model and the restricted regression model for normal data where the link function is not necessarily the identity link.

First consider n independent Poisson random variables with $E(Y_i) = \mu_i$. Thus $E(y) = \mu$ and $Cov(y) = D_\mu$ is a diagonal matrix with the elements of μ on the principal diagonal. Suppose that a log link is used, i.e. we have the model $\ln(\mu) = X\beta$ subject to constraints $L\beta = d$. In this case

$$h(\mu) = \ln(\mu) \text{ and } \frac{\partial h(\mu)}{\partial \mu} = \frac{\partial \ln(\mu)}{\partial \mu} = D_\mu^{-1}$$

From equation (4)

$$G_{\mu} = MD_{\mu}^{-1}$$

The ML estimate given in equation (2), becomes

$$\begin{aligned}\hat{\mu}_c &= y - (G_{\mu} D_{\mu})' (G_y D_{\mu} G_{\mu}')^* g(y) + o(\|y - \mu\|) \\ &= y - M' (MD_y^{-1} D_{\mu} D_{\mu}^{-1} M')^* g(y) + o(\|y - \mu\|) \\ &= y - M' (MD_y^{-1} M')^* g(y) + o(\|y - \mu\|),\end{aligned}$$

which is found by iterating over y . In this discussion the canonical link for the Poisson distribution, namely a log link, has been used, but any link function $h(\mu)$ may be used, as long as it is a monotonic differentiable function.

The other application is for $y \sim N(\mu, \sigma^2 I)$ where the model $\ln(\mu) = X\beta$ with $L\beta = d$ is entertained. Here the link function is not the canonical identity link, but the log link and estimating $\hat{\beta}_c$ will require an iterative procedure. As for the Poisson case, $\frac{\partial h(\mu)}{\partial \mu} = D_{\mu}^{-1}$ and $G_{\mu} = MD_{\mu}^{-1}$. The estimation procedure is given by

$$\begin{aligned}\hat{\mu}_c &= y - (G_{\mu} V_{\mu})' (G_y V_{\mu} G_{\mu}')^* g(y) + o(\|y - \mu\|) \\ &= y - (MD_{\mu}^{-1} \sigma^2 I)' (MD_y^{-1} \sigma^2 ID_{\mu}^{-1} M')^* g(y) + o(\|y - \mu\|) \\ &= y - (MD_{\mu}^{-1})' (MD_y^{-1} D_{\mu}^{-1} M')^* g(y) + o(\|y - \mu\|).\end{aligned}$$

Since D_{μ} is unknown, iteration takes place over both μ and y . (This double iterative procedure is explained by Matthews and Crowther (1995)).

8. Conclusion

The estimation procedure provides a simple method for finding the ML estimate, $\hat{\beta}_c$ in a generalized linear model in the presence of a set of linear restrictions $L\beta = d$. This is done by expressing the model, along with the

restrictions in the form of a function $g(\mu) = 0$, which satisfies the conditions of the estimation procedure of Proposition 1.

References

- AITCHISON, J. & SILVEY, S.D. (1958). Maximum-likelihood estimation of parameters subject to constraints. *Ann. Math. Statist.*, **29**, 813–828.
- HEWLETT, P.S. & PLACKETT, R.L. (1950). Statistical aspects of the independent joint action of poisons, particularly insecticides. II. Examination of data for agreement with the hypothesis. *Annals of Applied Biology*, **37**, 527–552.
- MATTHEWS, G.B. & CROWTHER, N.A.S. (1995). A maximum likelihood estimation procedure when modelling in terms of constraints. *South African Statist. J.*, **29**(1), 29–51.
- McCULLAGH, P. & NELDER, J.A. (1989) 3rd ed. *Generalized linear models*. London: Chapman & Hall.
- MONTGOMERY, D.C. & PECK, E.A. (1992) 2nd ed. *Introduction to linear regression analysis*. New York: John Wiley & Sons, Inc.
- NELDER, J. & WEDDERBURN, R.W.M. (1972). Generalized linear models. *J. Roy. Statist. Soc. A*, **135**, 370–384.
- NYQUIST, H. (1991). Restricted estimation of generalized linear models. *J. Appl. Statist.* **40**, No. 1, 133–141.
- SAS/STAT *User's guide, version 6, 4th ed.*, vol. 2. Cary NC: SAS Institute Inc., 1990.

Appendix

The SAS/IML program for Example 1 is as follows:

```
proc iml;
*Logistic regression with constraints;
reset nolog;
x={1 0 0 0.693147 0 0,
  1 0 0 0.97078 0 0,
  1 0 0 1.24703 0 0,
  1 0 0 1.52388 0 0,
  1 0 0 1.80171 0 0,
  1 0 0 2.07944 0 0,
  0 1 0 0 0.693147 0,
  0 1 0 0 0.97078 0,
  0 1 0 0 1.24703 0,
  0 1 0 0 1.52388 0,
  0 1 0 0 1.80171,
  0 1 0 0 2.07944 0,
  0 0 1 0 0 0.693147,
  0 0 1 0 0 0.97078,
  0 0 1 0 0 1.24703,
  0 0 1 0 0 1.52388,
  0 0 1 0 0 1.80171,
  0 0 1 0 0 2.07944};
yi={3,5,19,19,24,35,2,14,20,27,41,40,28,37,46,48,
  48,50};
ni={50,49,47,38,49,50,50,49,50,50,50,50,50,50,50,50,
```

```

50,50.000001});
L={ 0 0 0 0 0 1 -1,
    1 -1 0 0 0 0 0};
A=inv(X'*X)*X';
P=I(18)-x*inv(x'*x)*x';
e=j(18,1,1);
pi=yi/ni;
logit=log(pi/(e-pi));
B1=P*logit;
B2=L*A*logit;
B=B1//B2;
var=ni#pi#(e-pi);
ivar=1/var;
SIG=diag(var);
G1=P*(diag(ivar));
G2=L*A*(diag(ivar));
G=G1//G2;
m=yi;
diff=1;
itr=0;
do while (diff>0.00000001);
m1=m;
m=m-(G*SIG)*ginv(G*SIG*G')*B;
pi=m/ni;
logit=log(pi/(e-pi));
B1=P*logit;

```

```

B2=L*A*logit;
B=B1//B2;
var=ni#pi#(e-pi);
ivar=1/var;
SIG=diag(var);
G1=P*(diag(ivar));
G2=L*A*(diag(ivar));
G=G1//G2;
    diff=sqrt((m-m1)^(m-m1));
    itr=itr+1;
    end;
A=inv(x*x)*x;
Sc=SIG-(G*SIG)^ginv(G*SIG*G)*G*SIG;
H=inv(SIG);
VB=A*H*Sc*H*A;
print VB;
b=inv(x*x)*x*logit;
print "Estimated Regression Parameters";
print "";
print b [format=12.6];
print "";
print itr; print "";
print "Expected Frequencies and Observed Frequen-
cies";
print m yi;
*- - - Calculate deviance;

```

```
G2yi=2*yi`*log(yi/m);  
nmyi=ni-yi;  
nmyh=ni-m;  
g2nmyi=2*nmyi`*log(nmyi/nmyh);  
dev=G2yi+g2nmyi;  
print ````;  
print ``Deviance``;  
print dev;
```